

CloudFormation Console Full Guide (v2)

How to use AWS CloudFormation from the AWS Management Console

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1) What is CloudFormation?

AWS CloudFormation is an Infrastructure as Code (IaC) service that lets you define AWS resources in a template (JSON or YAML) and deploy them as a managed unit called a stack.

- Template: the code (JSON/YAML) describing resources.
- Stack: a deployed instance of a template.
- Parameters: runtime inputs to make templates reusable.
- Outputs: values returned after deployment (for reuse).
- Events: real-time deployment/update/delete logs.

2) CloudFormation Console Tour

- Stacks: list of all stacks in selected region.
- Events tab: detailed operation timeline and errors.
- Resources tab: actual created AWS resources.
- Outputs tab: exported values (IDs, ARNs, endpoints).
- Template tab: currently deployed template.
- Drift tab: configuration drift checks.
- Change sets: preview updates before execution.

3) Create a Stack in Console

Step-by-step

- Open AWS Console > search CloudFormation.
- Click Create stack > With new resources (standard).
- Specify template: Upload template file.
- Choose JSON file (example: 00-all-in-one.json).
- Enter stack name (example: student-all-in-one).
- Fill parameters (EnvironmentName, KeyPairName, etc.).
- Leave defaults for options unless your lab requires tags/permissions.
- Review and click Submit/Create stack.
- Monitor Events until status is CREATE_COMPLETE.

4) Validate Before Deploying

- Use CloudFormation template validation in Console (if available).
- Review JSON structure: commas, brackets, property names.
- Check parameter constraints and AllowedValues.
- Confirm region-specific resources exist (KeyPair, AMI, etc.).
- For S3, ensure generated bucket name will be globally unique.

5) Update an Existing Stack

- Open stack > click Update.
- Choose Replace current template or Use current template.
- Adjust parameters (e.g., instance type).
- Review and submit.
- Wait for UPDATE_COMPLETE.

Recommended teaching demo

- Change EC2 InstanceType from t3.micro to t3.small.

- Observe update behavior in Events.

6) Change Sets (Safe Update Preview)

Change sets let you preview what CloudFormation will modify before executing updates.

- From stack actions, create change set.
- Upload updated template / parameters.
- Review resource actions: Add / Modify / Replace / Remove.
- Execute change set only after review.

7) Drift Detection

Drift detection compares deployed stack resources with template definitions to find manual changes made outside CloudFormation.

- Open stack > Actions > Detect drift.
- Review drift status for each resource.
- Use drift findings for governance and remediation exercises.

8) Troubleshooting Failed Deployments

- Always check the first failed Event in red.
- Common issues:
 - - Missing permissions (IAM).
 - - Resource already exists (e.g., S3 bucket name conflict).
 - - Invalid parameter value.
 - - Regional mismatch (KeyPair not in region).
- Rollback behavior:
 - - CREATE_FAILED often rolls back automatically.
 - - You can inspect events to find exact failing resource.

9) Best Practices for Students

- Use Parameters for environment portability.
- Use Outputs to connect stacks.
- Tag every resource (Name, Environment, Owner).
- Prefer small modular templates for production.
- Use all-in-one templates for beginner labs and demos.
- Always clean up lab stacks to avoid charges.

10) Hands-On Classroom Lab Plan

Lab A - Beginner

- Deploy 00-all-in-one.json from console.
- Validate EC2 web page and S3 bucket settings.
- Delete stack and redeploy with new EnvironmentName.

Lab B - Intermediate

- Deploy network stack first (01-vpc-network.json).
- Deploy EC2 stack (02-ec2-instance.json) using imports.
- Deploy S3 stack (03-s3-bucket.json).
- Run change set and drift detection exercises.

11) Cleanup and Cost Control

- Delete in dependency order for modular labs:
 - 1) student-ec2
 - 2) student-s3
 - 3) student-network
- For all-in-one: delete student-all-in-one.

- Verify resources are deleted in EC2, VPC, and S3 consoles.

Optional CLI cleanup reference

```
aws cloudformation delete-stack --stack-name student-all-in-one
```

Prepared for Unixcloudtrainings by Valery